

Marcus D. Yang

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Education

University of Illinois Urbana-Champaign | Champaign, IL

August 2024 - May 2028

B.S. in Aerospace Engineering with a Computer Science minor, James Scholar Honors

GPA: 3.57/4

Relevant Coursework: Incompressible Flow, Aerospace Control Systems, Aerospace Structures, Thermodynamics, Engineering Materials, Linear Algebra, Electrical and Electronic Circuits, Intro to Computer Science II

Organizational and Technical Experience

TVC Rocket (3d Printing, Python)

Summer 2026

- Designed and fabricated a 2-axis thrust-vector-control gimbal, avionics bay, and recovery system by integrating a Teensy 4.1 flight computer, IMU, and servo actuators into a commercial rocket body..
- Simulated rocket descent dynamics in Python by sweeping sensor noise, motor burn variance, and wind disturbance across hundreds of Monte Carlo trials to validate a real-time landing-burn control law.
- Traced a servo-linkage resolution bottleneck in the TVC gimbal by characterizing the mechanical actuation ratio, then redesigned the control path to reach sub-degree pointing precision.
- Diagnosed two static-fire test failures on a thrust-vector-control gimbal by analyzing telemetry logs and frame-by-frame video, isolating a mis-referenced attitude axis and inverted control-loop signs.

Liquid Rocketry at Illinois

Spring 2025 - Present

- Modeled P&ID system architecture in Siemens NX and installed components into the pressurant tank assembly by sourcing, importing, and constraining CAD hardware to satisfy spatial constraints.
- Created CAD geometries for canard-grid fin configurations and ran ANSYS Fluent CFD cases to quantify aerodynamic interactions across 0–20° deflection.
- Assessed pressurant tank options, produced mechanical integration models (CAD) and verified fit against vehicle structural constraints and P&ID.

CAD Project (Siemens NX)

Spring 2025

- Recreated a full McDonnell F-101 Voodoo in Siemens NX by extracting and scaling geometric dimensions from multi-view engineering drawings to accurately reconstruct fuselage, wing, and tail geometry.
- Incorporated articulated control surfaces (ailerons, elevators, rudder) using parametric assemblies and constraints to render realistic aircraft deflection.

Glider Project

Fall 2024

- Built and tested gliders to optimize aerodynamic performance, achieving 101 ft glide distance and 3.9 s flight time, ranking 6th in class.
- Performed iterative flight testing and analysis of glide path, mass distribution, and wing geometry to maximize lift-to-drag performance and flight endurance.

Technical Skills

Computer Languages: Python, C++, Java, JavaScript

Software: Siemens NX, Fusion 360, Eagle PCB, KiCad, OpenRocket, XFLR5, Ansys Fluent, Excel

Shop Tools: Bambu A1 MiniSpot Welder, Belt Sander, Laser Cutter, CNC Plasma Cutter, Table Saw, Drill Press

Materials: Epoxy Resin, Fiberglass, Sheet Metal, Aluminum Pipes, Plywood, PETG, PLA

Leaderships and Activities

Asian American Association

Fall 2024 - Present

Treasurer | Fundraising Officer

- Managed membership operations for 300+ members, coordinating records and organizational logistics.
- Secured partnerships with 15 on-campus sponsors to support events and fundraising initiatives.

Flight Club Aerospace

Summer 2021 - Spring 2024

Landing Gear Research Lead | CAD | Fabrication

- Manufactured wing and control surfaces: cut foam ribs, trimmed aluminum spars with a circular saw, and applied epoxy & fiberglass layups to shape aerodynamic skins; aligned two 7-ft wings for flight testing.
- Drafted gussets and ribs in Fusion 360 and confirmed fit-checks between structural and aerodynamic components.
- Oversaw logistics for wing transport/disassembly/reassembly and assembled and presented landing-gear recommendations based on structural load and cost analyses.